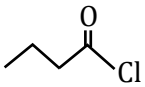
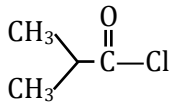
RACE Regular Analysis Through Continuous Exercise

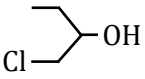
Organic Compounds Containing Nitrogen : RACE-01

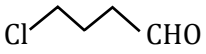
Reaction Involving Preparation of Oraganic Compound Containing Nitrogen

1. $(X)C_4H_7OCl \xrightarrow{NH_3} C_4H_9ON \xrightarrow[KOH]{Br_2} CH_3CH_2CH_2NH_2$; Compound (X) is :

(A) 

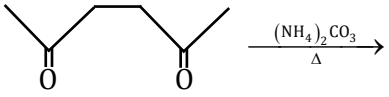
(B) 

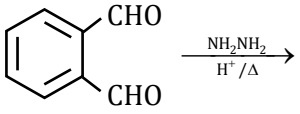
(C) 

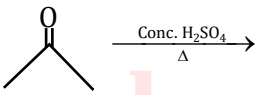
(D) 

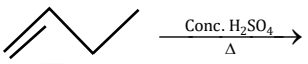
CCN150

2. Identify reactions that give aromatic products.

(A) 

(B) 

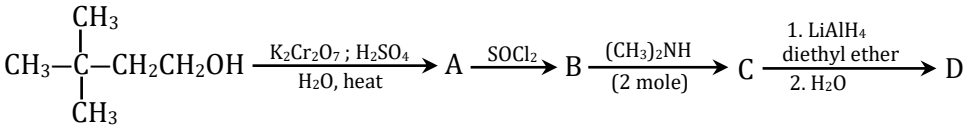
(C) 

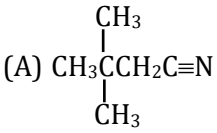
(D) 

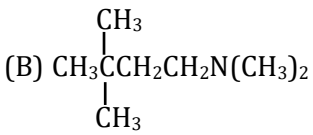
Rankers Academy JEE

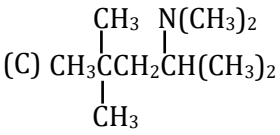
CCN151

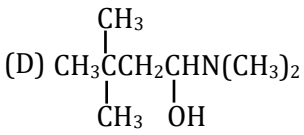
3. Identify product D in the following reaction sequence :



(A) 

(B) 

(C) 

(D) 

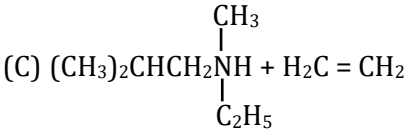
CCN153

4. The major products obtained from the following sequence of reactions are:

$(CH_3)_2CHCH_2N(CH_2CH_3)_2 \xrightarrow{CH_3I} \xrightarrow[H_2O]{Ag_2O} \xrightarrow{heat}$ produces

(A) $(CH_3)_2CHCH_2NH_2 + H_2C = CH_2$

(B) $(CH_3)_2NCH_2CH_3 + H_2C = C(CH_3)_2$

(C) 

(D) $(CH_3)_3\overset{+}{N}CH_2CH_3I^- + H_2C = CH_2$

CCN154

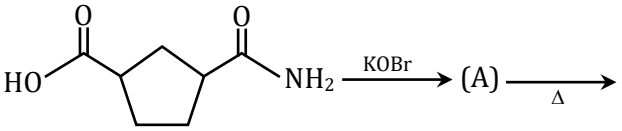
For More Join: @IITJEE_Advanced

5. A nitrile X is treated with $LiAlH_4$ to obtain compound Y(C_2H_7N). In a separate reaction X is hydrolyzed in an acid medium to obtain Z. The product obtained after mixing Y and Z will be?

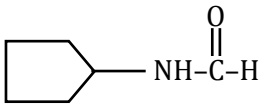
(A) $CH_3CONHCH_2CH_3$ (B) $CH_3CH_2CONHCH_2CH_3$

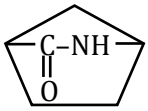
(C) $(CH_3COO^-)(CH_3CH_2NH_3^+)$ (D) $(CH_3CH_2COO^-)(CH_3NH_2^+)$

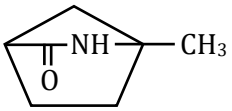
CCN156

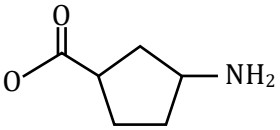
6. 

(B). Compound (B) is :

(A) 

(B) 

(C) 

(D) 

CCN158

7. Which of the following will no give Hoffmann bromamide reaction ?

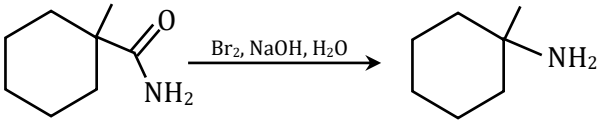
(A) $CH_3-\overset{\overset{O}{\|}}{C}-NH_2$ (B) $Ph-\overset{\overset{O}{\|}}{C}-NH_2$

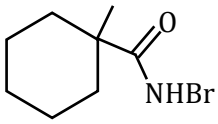
(C) $CH_3-\overset{\overset{O}{\|}}{C}-NH-Br$ (D) $Ph-\overset{\overset{O}{\|}}{C}-NH-Ph$

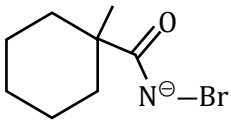
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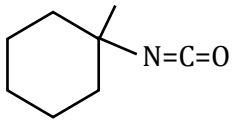
CCN159

8. Which of the following species would no be involved in the Hoffmann rearrangement shown below ?



(A) 

(B) 

(C) 

(D) All of the above are involved in the reaction.

CCN160

9. $CH_3CH_2-CN \xrightarrow[H_2O]{LiAlH_4} X$

$CH_3CH_2-NC \xrightarrow[H_2O]{LiAlH_4} Y$

Identify the relation between product X & Y.

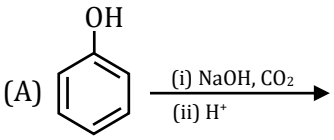
(A) Stereoisomers (B) Functional group isomers

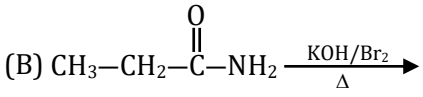
(C) Positional isomers (D) Chain isomers

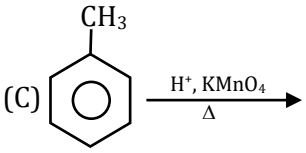
CCN191

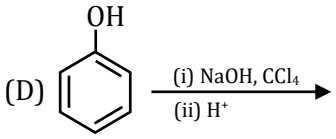
For More Join: @IITJEE_Advanced

10. Which of the following has –COOH group in major product ?

(A) 

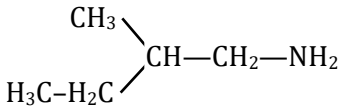
(B) 

(C) 

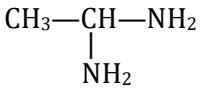
(D) 

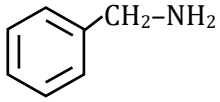
CCN192

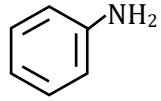
11. The total number of amines among following which can be synthesize by Gabriel synthesis

(a) 

(b) $CH_3–CH_2–NH_2$

(c) 

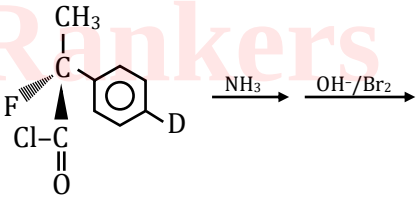
(d) 

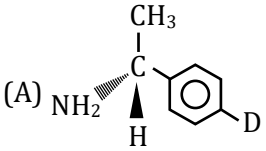
(e) 

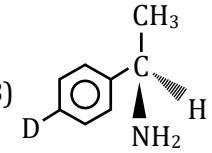
(f) $CH_3–NH–CH_3$

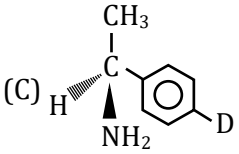
CCN193

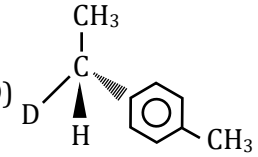
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12. 

(A) 

(B) 

(C) 

(D) 

CCN194

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	ABC	B	C	C	B	D	D	B	ACD
Question	11	12								
Answer	4	C								

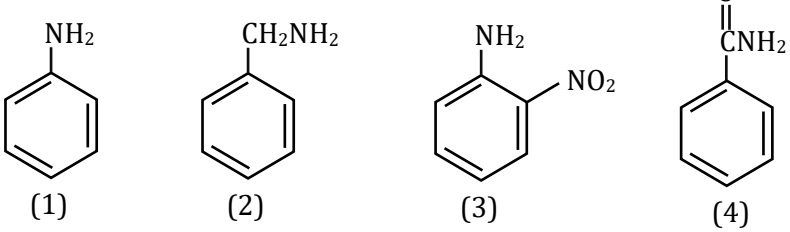
For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Organic Compounds Containing Nitrogen : RACE -02

Reaction Involving Oraganic Compound Containing Nitrogen

1. Rank he following compounds in order of increasing basic strength. (weakest → strongest):



(1) (2) (3) (4)

(A) 4 < 2 < 1 < 3 (B) 4 < 3 < 1 < 2 (C) 4 < 1 < 3 < 2 (D) 2 < 1 < 3 < 4

CCN152

2. $CH_3CH_2COOH \xrightarrow{NH_3} \xrightarrow{\Delta} \xrightarrow{Br_2/KOH} (A)$

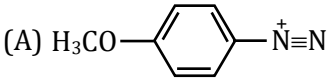
Which is not correct statement for (A)

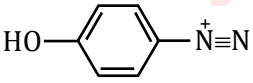
(A) It gives carbyl amine reaction. (B) It gives mustard oil reaction.

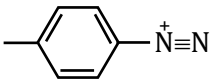
(C) Forms alcohol with $NaNO_2$/ HCl (D) Forms yellow oily liquid with $NaNO_2$/ HCl

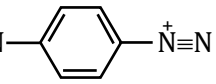
CCN195

3. Which of the following is most reactive ion in azocoupling reaction ?

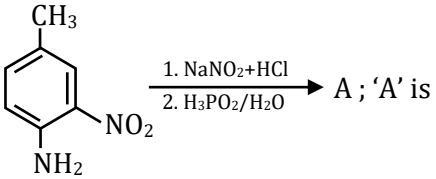
(A) 

(B) 

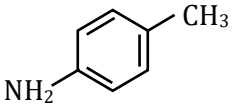
(C) 

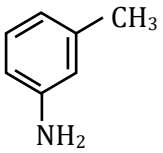
(D) 

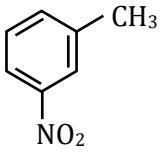
CCN196

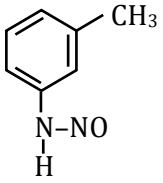
4. 

A ; 'A' is

(A) 

(B) 

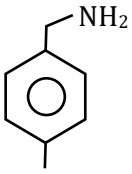
(C) 

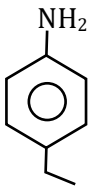
(D) 

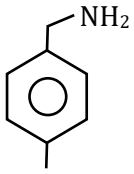
CCN197

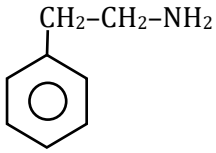
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5. Three bottles A, B and C have been found, each of which contains a liquid and is labeled "amine $C_8H_{11}N$." As an expert in amine chemistry, you have been hired as a consultant and asked to identify each compound. Compounds A and B give off a gas when they react with $NaNO_2$ and HCl at 0°C; C does not. However, when the aqueous reaction mixture from the diazotization of C is warmed, a gas is evolved. Compound A is optically inactive. Oxidation of C with H_2O_2 (a reagent known to oxidize amino groups to nitro groups), followed by vigorous oxidation with $KMnO_4$ gives p–nitrobenzoic acid. Oxidation of B in a similar manner yields 1,4–benzene–dicarboxylic acid (terephthalic acid), and oxidation of A yields benzoic acid. Identify compounds A, B and C.

(A) A is 

(B) C is 

(C) B is 

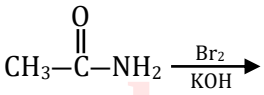
(D) A is 

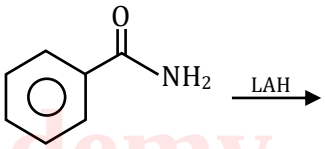
CCN198

6. What is the number of benzene derivatives having the molecular formula C_7H_9N that would respond positive to carbylamine test.

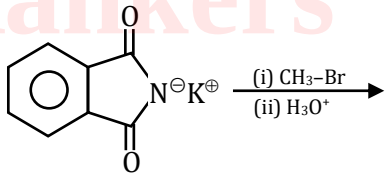
CCN199

7. In which of the following reaction, one of the product obtained will give Hoffmann mustard oil test?

(A) 

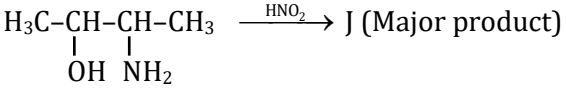
(B) 

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(C) 

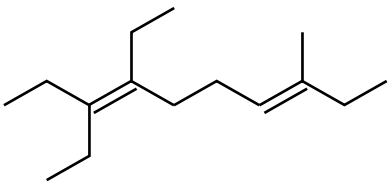
(D) All of these

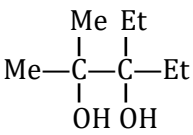
CCN201

8. 

Which statement is incorrect about above major product ?

(A) It can be prepared by reaction of But-2-yne with Hg^{2+} / H^+ / H_2O

(B) It can be prepared by ozonolysis of 

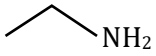
(C) J on reaction with Mg–Hg followed by acidification gives  as one of the product.

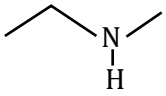
(D) It can form an oxime on reaction with NH_2–OH which shows geometrical isomerism.

CCN202

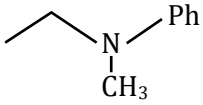
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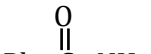
9. Which of the following will show positive carbyl amine test ?

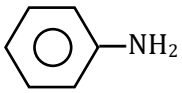
(1) 

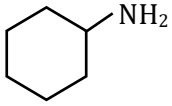
(2) 

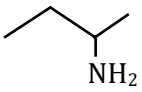
(3) $Ph-CH_2-NH_2$

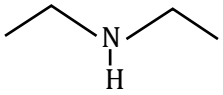
(4) 

(5) 

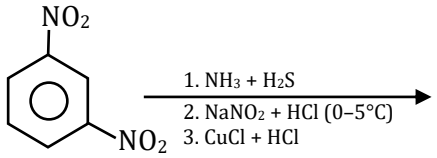
(6) 

(7) 

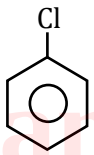
(8) 

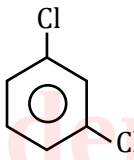
(9) Ph 

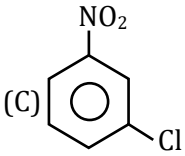
CCN203

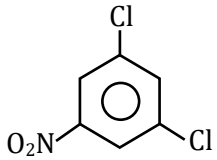
10. 

Final Product is?

(A) 

(B) 

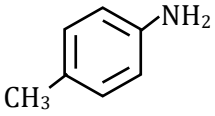
(C) 

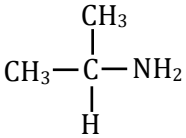
(D) 

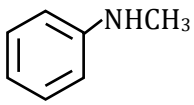
CCN204

11. Which one of the following gives the most stable Diazonium salt ?

(A) $CH_3-CH_2-CH_2-NH_2$

(B) 

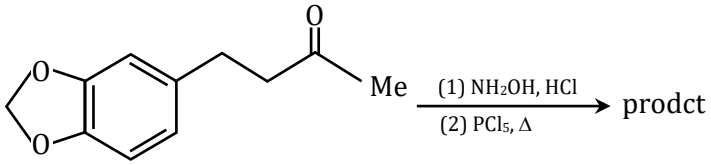
(C) 

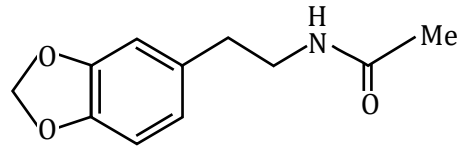
(D) 

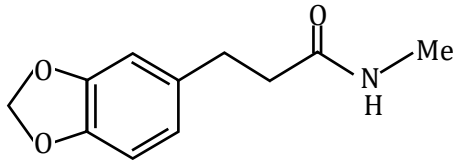
CCN205

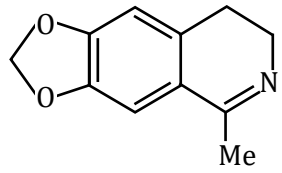
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12. Which of the following product will be obtained in the given (consider minor product also) Beckmann-type rearrangement?



(A) 

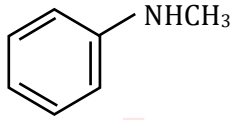
(B) 

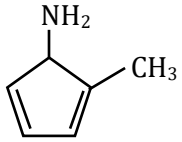
(C) 

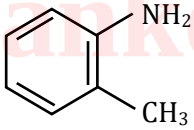
(D) All of these

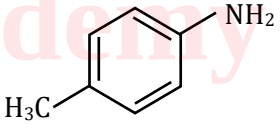
CCN155

13. The compound X(C_7H_9N) reacts with benzensulfonyl chloride to give Y($C_{13}H_{13}NO_2S$) which is insoluble in alkali. The compound X is -

(A) 

(B) 

(C) 

(D) 

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CCN157

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	D	D	C	BCD	4	D	C	5	C
Question	11	12	13							
Answer	B	D	A							

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